

NICHOLAS RUFFIN

Data Engineering Leader | Enterprise PHI/PII Pipelines & Regulatory Compliance

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PROFESSIONAL SUMMARY

Enterprise data engineering leader with 12+ years of experience building and operating large-scale pipelines in highly regulated healthcare environments. Expert in PHI/PII protection, de-identification, and tokenization at enterprise scale — currently processing 4.8B pharmacy records through HIPAA-compliant pipelines with strict regulatory controls. Proven director-level leadership of 16+ engineers across cross-functional teams, with deep experience in healthcare claims data, regulatory compliance, and AI/ML-ready data infrastructure. Combines hands-on technical expertise in Snowflake, AWS, and Python with the organizational leadership required to drive outcomes across large, matrixed environments.

TECHNICAL SKILLS

Data Platforms	Snowflake (Procs, Streams, Tasks, Multi-instance), AWS (MWAA, S3, Lambda, Fargate, IAM), Kafka, Data Warehousing
Compliance	HIPAA, PHI/PII De-identification, Datavant Tokenization, Expert Determination, Audit Readiness, Access Controls
Engineering	Python, SQL, ETL/ELT Pipeline Design, Enterprise Data Architecture, AI/ML Data Readiness
Leadership	Managing managers, team building (16+ across US/India), cross-functional partnership, vendor evaluation, technical strategy
DevOps	Apache Airflow, Terraform, GitHub Actions (CI/CD), Docker, Git

EXPERIENCE

Manager, Data Product Engineering | RedSail Technologies

Feb 2025 – Present

Promoted from Senior Clinical Data Engineer. Leading the data monetization engineering team.

- Own end-to-end enterprise data platform generating **\$7M+ ARR**, architecting secure de-identification and tokenization pipelines (Snowflake + Datavant) processing **4.8 billion PHI records** with strict regulatory compliance controls.
- Oversee regulatory compliance including HIPAA audit readiness, proof-of-delivery documentation, Expert Determination coordination, and PHI/PII protection governance across all pipeline operations.
- Architected next-generation V2 tokenization pipeline (AWS Fargate + Snowflake), designing enterprise-scale de-identification workflows and automated orchestration spanning **50+ daily production pipelines**.
- Lead cross-functional collaboration with product management, infrastructure engineering, compliance teams, and external vendor partners to drive platform roadmap in a regulated environment.
- Manage client delivery relationships across **5+ pharmaceutical data buyers**, owning delivery SLAs, schema specifications, and data quality standards at enterprise scale.
- Building and onboarding data engineering team, establishing technical standards, code review processes, and deployment workflows via Terraform and GitHub Actions.

Director of Full Stack Development | Nimble Solutions

Aug 2021 – Jan 2025

Promoted from Lead Software Engineer to Director by the CIO. Built and managed an engineering department of 16+ people across three functional teams: 5 US software engineers, 6 SQL/data developers, 5 offshore developers in Hyderabad, and a project manager.

- Led technology due diligence for **4 acquisitions** in regulated healthcare — evaluating data security, compliance posture, and integration complexity. Traveled internationally to integrate acquired teams.
- Partnered with CIO, VP of Infrastructure, and CISO on **technical strategy, vendor evaluations, and TCO analysis** for enterprise healthcare systems in regulated environments.
- Engineered integration of **50+ data sources** into a unified enterprise workflow, designing API orchestration patterns for high reliability across large-scale operations.
- Designed and led the build of **QUARC**, a queue-based claims reimbursement system ordered by revenue with historical AR aging data, increasing throughput to **30+ claims/day per rep.**
- Established technical standards, implemented CI/CD pipelines reducing deployment time by **60%**, and achieved **90%+ test code coverage** across enterprise applications.
- Mentored engineers across all three teams, managing both ICs and team leads in a cross-functional, matrixed environment.

Senior Full Stack Developer | Core & Main

Jun 2017 – Aug 2021

- Developed enterprise data applications using C#/.NET and SQL, supporting **200+ concurrent** corporate users across operations.

EDUCATION

Bachelor of Science, Information Technology

University of Phoenix